Magic Number: **115952**

Q1. As the results of this exercise provide the magic number, provide the code you execute and screen shots of the following commands:

- foodratings.printSchema()
- foodratings.show(5)

Code Executed

```
from pyspark.sql.types import StructType, StructField, StringType, IntegerType
from pyspark.sql.functions import col, sum as _sum
import hashlib

schema_ratings = StructType([
    StructField("name", StringType(), True),
    StructField("food1", IntegerType(), True),
    StructField("food2", IntegerType(), True),
    StructField("food3", IntegerType(), True),
    StructField("food4", IntegerType(), True),
    StructField("placeid", IntegerType(), True),
])

foodratings = (spark.read
    .option("header","false")
    .schema(schema_ratings)
    .csv("hdfs:///user/hadoop/foodratings.txt"))

foodratings.printSchema()
foodratings.show(5, truncate=False)

# MAGIC NUMBERS
row_count = foodratings.count()

sums_row = (foodratings
    .select(_sum(col("food1")).alias("sum_food1"),
            _sum(col("food2")).alias("sum_food2"),
            _sum(col("food3")).alias("sum_food3"),
            _sum(col("food4")).alias("sum_food4"))
    .collect()[0])

checksum_src = f"{row_count}-{sums_row['sum_food1']}-{sums_row['sum_food2']}-{sums_row
    ['sum_food3']}-{sums_row['sum_food4']}"

magic_checksum = hashlib.md5(checksum_src.encode("utf-8")).hexdigest()

print("Row Count Magic Number:", row_count)
print("Column Sums:", dict(sums_row.asDict()))
print("MD5 Checksum:", magic_checksum, f"(from '{checksum_src}')")
```

```
AWS Keys — hadoop@ip-10-0-0-202:~ — ssh -i ~/Downloads/AWS Keys/emr-key-pair.pem hadoop@...
[hadoop@ip-10-0-0-202 ~]$ pyspark
Python 3.9.25 (main, Dec 10 2025, 00:00:00)
[GCC 11.5.0 20240719 (Red Hat 11.5.0-5)] on linux
Type "help", "copyright", "credits" or "license" for more information.
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
26/03/05 03:35:44 WARN HiveConf: HiveConf of name hive.server2.thrift.url does not exist
26/03/05 03:35:46 WARN Client: Neither spark.yarn.jars nor spark.yarn.archive is set, falling back to upl
oading libraries under SPARK_HOME.
Welcome to
      ____              __
     / __/__  ___ _____/ /__
    _\ \/ _ \/ _ `/ __/  '_/
   /__ / .__/\_,_/_/ /_/\_\   version 3.5.6-amzn-1
      /_/

Using Python version 3.9.25 (main, Dec 10 2025 00:00:00)
Spark context Web UI available at http://ip-10-0-0-202.ec2.internal:4040
Spark context available as 'sc' (master = yarn, app id = application_1772678645065_0004).
SparkSession available as 'spark'.
>>> exec(open("/home/hadoop/assignment7_solution.py").read())

=== Using paths ===
FOODRATINGS_PATH: hdfs:///user/hadoop/foodratings115952.txt
FOODPLACES_PATH : hdfs:///user/hadoop/foodplaces115952.txt

--- Exercise 1: foodratings.printSchema() ---
root
 |-- name: string (nullable = true)
 |-- food1: integer (nullable = true)
 |-- food2: integer (nullable = true)
 |-- food3: integer (nullable = true)
 |-- food4: integer (nullable = true)
 |-- placeid: integer (nullable = true)


--- Exercise 1: foodratings.show(5) ---
+----+-----+-----+-----+-----+-------+
|name|food1|food2|food3|food4|placeid|
+----+-----+-----+-----+-----+-------+
|Joe |18   |22   |11   |27   |3      |
|Mel |39   |33   |29   |41   |2      |
|Joe |34   |43   |7    |30   |3      |
|Mel |17   |26   |42   |34   |5      |
|Jill|39   |24   |42   |49   |3      |
+----+-----+-----+-----+-----+-------+
only showing top 5 rows


--- Exercise 1: MAGIC NUMBERS ---
Row Count Magic Number: 1000
Column Sums: {'sum_food1': 26096, 'sum_food2': 25280, 'sum_food3': 24685, 'sum_food4': 25091}
MD5 Checksum: ce1c1a6981bcac898fca7a5286ce06c1 (from '1000-26096-25280-24685-25091')
```

Q2. Load the 'foodplaces' file as a 'csv' file into a DataFrame called foodplaces. When doing so specify a schema having fields of the following names and types:

Field Name	Field Type
placeid	Integer
placename	String

As the results of this exercise provide the code you execute and screen shots of the following commands:

- foodplaces.printSchema()
- foodplaces.show(5)

```
schema_places = StructType([
    StructField("placeid", IntegerType(), True),
    StructField("placename", StringType(), True),
])

foodplaces = (spark.read
    .option("header","false")
    .schema(schema_places)
    .csv("hdfs:///user/hadoop/foodplaces.txt"))

foodplaces.printSchema()
foodplaces.show(5, truncate=False)
```

Q3. Step A

Register the DataFrames created in exercise 1 and 2 as tables called "foodratingsT" and "foodplacesT"

```
foodratings.registerTempTable('foodratingsT')
foodplaces.registerTempTable('foodplacesT')
```

Step B

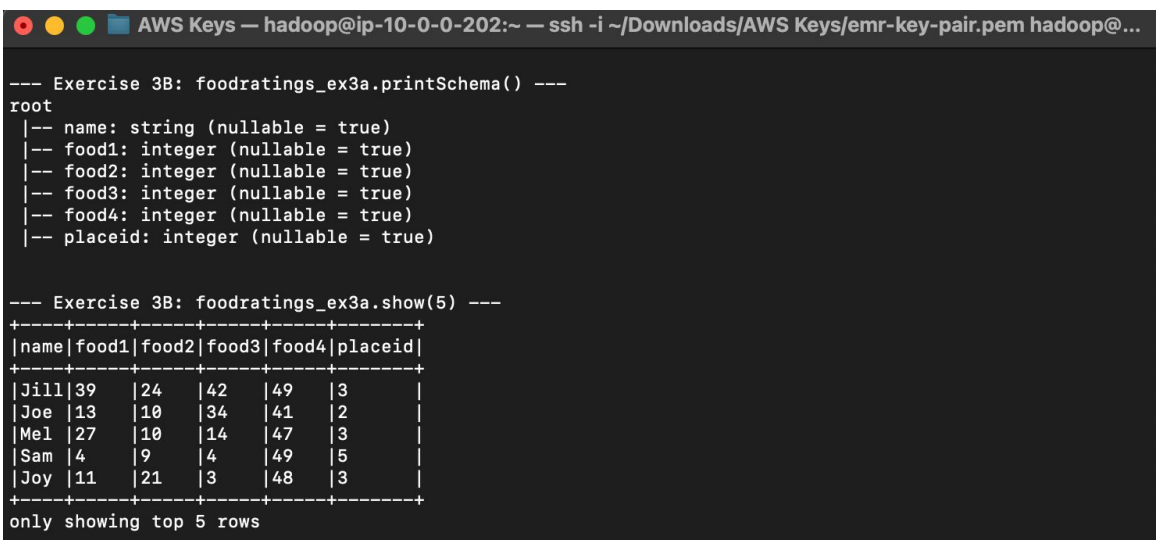
Use a SQL query on the table "foodratingsT" to create a new DataFrame called foodratings_ex3a holding records which meet the following condition: food2 < 25 and food4 > 40. Remember, when defining conditions in your code use maximum parentheses.

As the results of this step provide the code you execute and screen shots of the following commands:

- foodratings_ex3a.printSchema()
- foodrating_ex3a.show(5)

```
# 3B: food2 < 25 AND food4 > 40
foodratings_ex3a = spark.sql("""
    SELECT * FROM foodratingsT
    WHERE (food2 < 25) AND (food4 > 40)
""")

foodratings_ex3a.printSchema()
foodratings_ex3a.show(5, truncate=False)
```



Step C

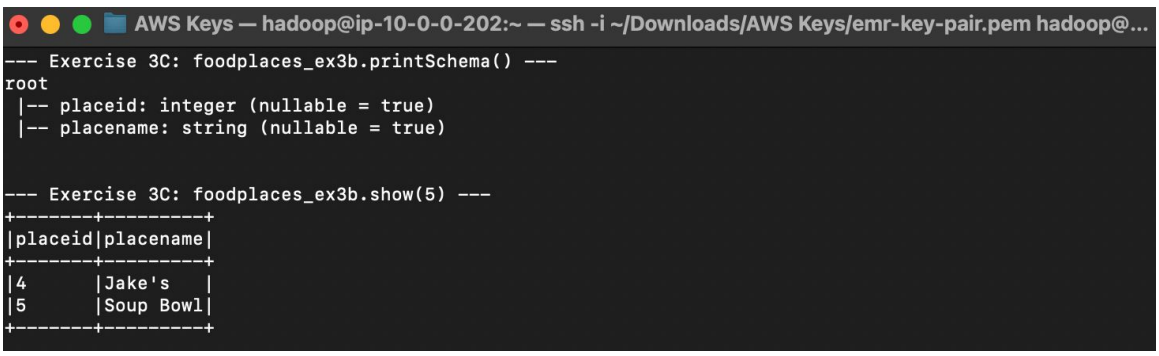
Use a SQL query on the table "foodplacesT" to create a new DataFrame called foodplaces_ex3b holding records which meet the following condition: placeid > 3.

As the results of this step provide the code you execute and screen shots of the following commands:

- foodplaces_ex3b.printSchema()
- foodplaces_ex3b.show(5)

```
# 3C: placeid > 3
foodplaces_ex3b = spark.sql("""
    SELECT * FROM foodplacesT
    WHERE (placeid > 3)
""")

foodplaces_ex3b.printSchema()
foodplaces_ex3b.show(5, truncate=False)
```



Q4. Use a transformation (not a SparkSQL query) on the DataFrame 'foodratings' created in exercise 1 to create a new DataFrame called foodratings_ex4 that includes only those records (rows) where the 'name' field is "Mel" and food3 < 25.

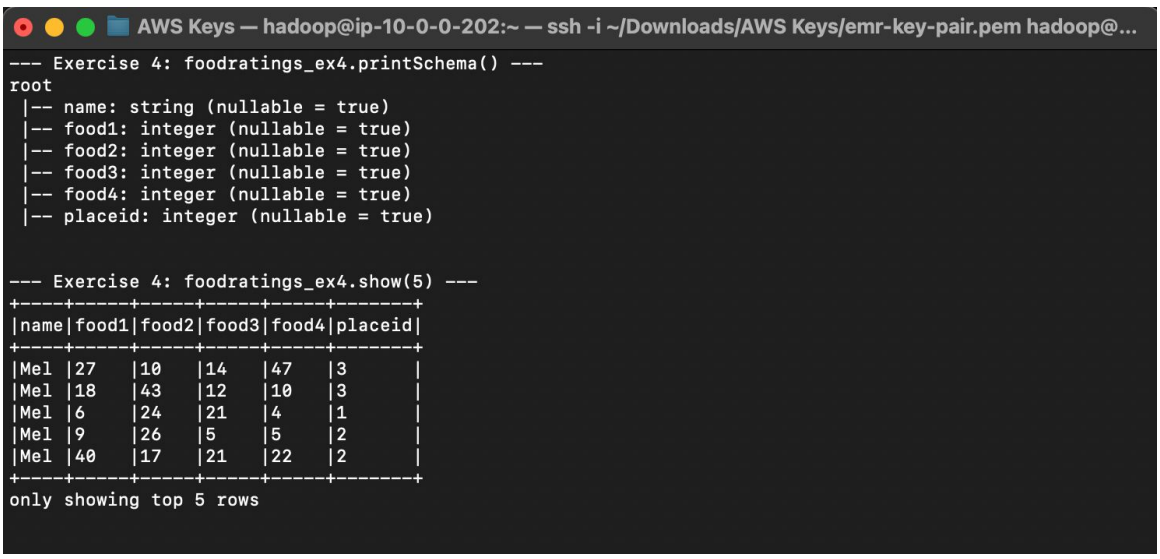
As the results of this step provide the code you execute and screen shots of the following commands:

- foodratings_ex4.printSchema()
- foodratings_ex4.show(5)

```
from pyspark.sql.functions import col

foodratings_ex4 = foodratings.filter(
    (col("name") == "Mel") & (col("food3") < 25)
)

foodratings_ex4.printSchema()
foodratings_ex4.show(5, truncate=False)
```



Q5. Use a transformation (not a SparkSQL query) on the DataFrame 'foodratings' created in exercise 1 to create a new DataFrame called foodratings_ex5 that includes only the columns (fields) 'name' and 'placeid'

As the results of this step provide the code you execute and screen shots of the following commands:

- foodratings_ex5.printSchema()
- foodratings_ex5.show(5)

```
foodratings_ex5 = foodratings.select("name", "placeid")

foodratings_ex5.printSchema()
foodratings_ex5.show(5, truncate=False)
```

```
--- Exercise 5: foodratings_ex5.printSchema() ---
root
 |-- name: string (nullable = true)
 |-- placeid: integer (nullable = true)


--- Exercise 5: foodratings_ex5.show(5) ---
+----+-------+
|name|placeid|
+----+-------+
|Joe |3      |
|Mel |2      |
|Joe |3      |
|Mel |5      |
|Jill|3      |
+----+-------+
only showing top 5 rows
```

Q6. Use a transformation (not a SparkSQL query) to create a new DataFrame called ex6 which is the inner join, on placeid, of the DataFrames 'foodratings' and 'foodplaces' created in exercises 1 and 2

As the results of this step provide the code you execute and screen shots of the following commands:

- ex6.printSchema()
- ex6.show(5)

```
ex6 = foodratings.join(foodplaces, on="placeid", how="inner")

ex6.printSchema()
ex6.show(5, truncate=False)
```

```
--- Exercise 6: ex6.printSchema() ---
root
 |-- placeid: integer (nullable = true)
 |-- name: string (nullable = true)
 |-- food1: integer (nullable = true)
 |-- food2: integer (nullable = true)
 |-- food3: integer (nullable = true)
 |-- food4: integer (nullable = true)
 |-- placename: string (nullable = true)


--- Exercise 6: ex6.show(5) ---
+-------+----+-----+-----+-----+-----+---------+
|placeid|name|food1|food2|food3|food4|placename|
+-------+----+-----+-----+-----+-----+---------+
|3      |Joe |18   |22   |11   |27   |Food Town|
|2      |Mel |39   |33   |29   |41   |Atlantic |
|3      |Joe |34   |43   |7    |30   |Food Town|
|5      |Mel |17   |26   |42   |34   |Soup Bowl|
|3      |Jill|39   |24   |42   |49   |Food Town|
+-------+----+-----+-----+-----+-----+---------+
only showing top 5 rows

>>> 
```